Elias Hossain

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Education

University of Central Florida — Orlando, FL
Ph.D. in Industrial Engineering (CS concentration), GPA: 3.87/4.00 — Aug 2025 – 2028 (Expected)

Mississippi State University — Mississippi State, MS
M.S. in Computer Science (Research), GPA: 3.88/4.00 — Jan 2024 – Jul 2025

Technical and Research Experience

Graduate Research Assistant — *Aug 2025 – Present*
Complex Adaptive Systems Laboratory, University of Central Florida

- Designed **NEXUS**, a runtime safety monitor for tool-using LLM agents, improving intervention accuracy by **27.3 percentage points** over rule-only monitoring with **0.949 F1** and **0.205 ms** median latency.
- Developed **ChannelGuard**, training-free communication gates for multi-agent LLMs; evaluated **2,100 traces** across **8 attack families** and **3 model backends**, eliminating tool-output poisoning in the reported evaluation.
- Built multi-agent preference-data verification methods, reducing Qwen2-7B summarization refusals from **14.8% to 0.5%** while maintaining win rate.
- Released **5 public ML benchmarks (74,590+ records)** and reference implementations; performed multi-GPU experiments on Qwen, Llama, and Mistral models.

Graduate Research Assistant — *May 2025 – Jul 2025*
Mississippi State University

- Developed contrastive-learning and uncertainty-estimation approaches for detecting rare malicious events in imbalanced security logs.
- Implemented **FAISS** retrieval and **RAGAS** evaluation for BarkPlug V.2, a campus-deployed retrieval-augmented assistant.

Software Engineer — *Dec 2022 – Nov 2023*
REVE Systems, Dhaka, Bangladesh

- Improved speech-recognition **word error rate by 20%** and reduced **inference time by 15%** using DeepSpeech and Wav2Vec 2.0; deployed models in a production e-governance voice interface.

AI/ML Researcher — *Oct 2020 – Nov 2022*
Time Research & Innovation, UK

- Developed X-ray/CT diagnostic models for COVID-19 screening and ML components for a telemedicine platform.

Selected Technical Projects

MiniHealthLM: Transformer Language Model from Scratch | GitHub | *Python, PyTorch, GQA, RoPE*

- Built a **600M-parameter** Qwen-style Transformer with grouped-query attention, RoPE, and RMSNorm; developed medical-text pretraining with PII filtering, a custom BPE tokenizer, and checkpointing.

MLDriftGuard: ML Data Drift Detection Package | GitHub | *Python, SciPy, pandas*

- Developed an installable Python package using **KS and chi-squared tests** for numerical/categorical drift, with visualizations and automated mitigation suggestions.

Selected Publications & Preprints

- **Hossain et al.** "When Policies Cannot Be Retrained: A Unified Closed-Form View of Post-Training Steering in Offline Reinforcement Learning." *NeurIPS 2026 (main track).* [arXiv]
- **Hossain et al.** "NEXUS: Structured Runtime Safety for Tool-Using LLM Agents." *Findings of AACL-IJCNLP 2026.* [arXiv]
- **Hossain et al.** "Right Knowledge, Wrong Answer: Test-Time Steering for Temporal Fact Conflicts in Open-Weight Language Models." *Under review, AAAI 2027.* [arXiv]
- **Hossain et al.** "ChannelGuard: Safe Models Do Not Compose into Safe Multi-Agent Systems." *Under review, ACL ARR.* [arXiv]
- **Hossain et al.** "UAT-LITE: Inference-Time Uncertainty-Aware Attention for Pretrained Transformers. *Committed to EACL 2027.* [arXiv]

Technical Skills

Languages: Python, SQL, R
ML / AI: PyTorch, Hugging Face Transformers, scikit-learn, RLHF, DPO, SFT, LoRA/QLoRA, offline RL, LLM evaluation, RAG, FAISS, multi-agent systems
Tools / Platforms: Git, Docker, Azure, Google Cloud Platform (GCP), multi-GPU training and evaluation